College of Computing and Information Sciences
NORTHWEST SAMAR STATE UNIVERSITY
Resilience • Integrity • Service • Excellence

PATHFINDER: MOBILE - BASED 3D NAVIGATION SYSTEM

A Capstone Project
Presented to

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Calbayog City

In Partial Fulfillment of the
Requirements for the Degree
BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY

By

MARY ROSE E. LEBARIO

JOHN ARNAIS M. RONDINA

ROBERT JOHN L. LABONITE

JODELYN E. MONTERNEL

VIOSEL D. LOJERO

MAY 2026

APPROVAL SHEET

The Capstone Project entitled "**PATHFINDER: MOBILE – BASED 3D NAVIGATION SYSTEM**" prepared and submitted by **MARY ROSE E. LEBARIO, JODELYN E. MONERNEL, VIOSEL D. LOJERO, JOHN ARNAIS M. RONDINA,** and **ROBERT JOHN L. LABONITE** in partial fulfillment of the course requirements for the Degree of Bachelor of Science in Information Technology, is hereby examined and recommended for acceptance and approval.

CAPSTONE PROJECT COMMITTEE

CRISTY S. CARPON, MSIT
Member

AARON JUDE C. PAEL
Member

RONALD A. AMOGUEZ
Member

JOSHUA C. BANZON
Capstone Adviser

Accepted and approved in partial fulfillment of the requirements for the Degree of Bachelor of Science in Information Technology.

FRANCISCO B. BACAMANTE JR., DIT
Dean, College of Computing and Information Sciences

APPROVAL

APPROVED by the tribunal of the Oral Examination with the grade of _____.

PANEL OF EXAMINERS

CRISTY S. CARPON, MSIT
Member

AARON JUDE C. PAEL
Member

RONALD A. AMOGUEZ
Member

JOSHUA C. BANZON
Capstone Adviser

FRANCISCO B. BACAMANTE JR., DIT
Dean, College of Computing and Information Sciences

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TEAM PATHFINDER

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EXECUTIVE SUMMARY

Navigating within school campuses can often be confusing for newcomers, visitors, and regular staff member, especially when directions depend on printed maps or signboard. To address this challenge, the researchers developed **PathFinder: Mobile - Based 3D Navigation System**, a mobile application designed to enhance wayfinding and accessibility with the College Computing and Information Sciences.

The main objective of the project is to provide a reliable, user- friendly, and interactive solution for locating classrooms, offices, laboratories and other essential areas. By offering an innovative 3D navigation system, the study aims to reduce confusion, save time, and create more convenient campus experience for student, faculty, and visitors.

The scope of the application includes major features such as a detailed 3D model of campus building, and intuitive search bar for locating rooms or facilities, and offline functionality to ensure access even without an internet connection. The system also provides step by step navigation that helps users reach their destination. The team adopted the Rapid Application Development Methodology, which emphasizes iterative design, rapid prototyping, and integration of user feedback. Visual Studio was used as the primary integrated development environment for coding and debugging; Blender for creating realistic 3D models of the campus, and Capacitor, a cross-platform runtime framework, to integrate web-based code into the mobile environment and ensure smooth performance on Android devices.

The significance of the study lies in its contribution to Northwest Samar State University as an institution. The system not only improves accessibility and campus orientation but also demonstrates the application of modern information technology in enhancing academic environments. Ultimately, PathFinder promotes efficiency, convenience, and innovation, making campus navigation simpler, more engaging, and more accessible for everyone.

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CHAPTER 1

INTRODUCTION

Project Context

Navigating large university campuses has long posed a challenge for many individuals, particularly freshmen, visitors, and newly hired faculty members. Studies have shown that first-time campus visitors often experience confusion and stress when trying to locate specific classrooms, laboratories, and offices due to the complex layout of university grounds Al-Ahmad & Al-Azizi (2020). The absence of a centralized navigation system can lead to delays and frustration, especially during important academic periods such as enrollment or examinations Chen et al (2019). Traditional methods of campus orientation such as printed maps, static signage, or verbal directions are often inefficient and may fail to provide accurate, real-time guidance in large or crowded environments Wang & Lee (2021). As stated by Tahir and Krogstie (2023), “inadequate navigation tools increase users’ mental workload and significantly reduce user satisfaction, especially in complex indoor and outdoor environments,” emphasizing the need for more efficient and user-friendly navigation solutions in educational institutions.

Universities in other countries have begun integrating Augmented Reality (AR), Wi-Fi, or Global Positioning System (GPS) enabled indoor navigation systems. However, these approaches typically require expensive infrastructure that is not feasible for most educational institutions in the Philippines. As educational spaces grow more complex, there is an increasing need for innovative and cost-effective solutions that support easy, accessible, and interactive

navigation within school environments. Sanchez (2023) and Tiwari & Gupta (2018) supports the use of 3D visualization tools to enhance spatial awareness and ease, better understand their surroundings and move efficiently from one place to another.

The College of Computing and Information Sciences at Northwest Samar State University is one of the university's most active departments, frequently visited by students, faculty, and guests. Despite its importance, the lack of an organized navigation system often leads to confusion when locating classrooms, laboratories, or faculty offices. The current setup offers no centralized digital map or guidance tool, which hampers convenience and accessibility. Without a proper navigation system, individuals unfamiliar with the campus may waste valuable time trying to find their destinations, resulting in frustration and decreased efficiency.

To address this, the proposed project introduces Pathfinder, a mobile-based 3D Navigation System specifically designed for College of Computing and Information Sciences. The app will feature a rendered 3D model of the building and its facilities, allowing users to interactively explore the layout, search for specific locations, and receive basic route guidance. Built using the Unity Engine and 3D modeling tools, Pathfinder will provide an intuitive interface that is accessible even to users with limited tech experience. The application will function both online and offline, ensuring reliability at all times. Ultimately, Pathfinder aims to improve campus mobility, minimize time spent searching for locations, and provide a modern and efficient navigation experience for the Northwest Samar State University community.

Purpose and Description

The purpose of this project is to develop a mobile application that will serve as a 3D navigation tool specifically for the College of Computing and Information Sciences at Northwest Samar State University. The application aims to assist students, faculty, and visitors in locating lecture rooms, offices, laboratories, and other key facilities within the CCIS building. Through an intuitive and user-friendly interface, the app will provide an interactive 3D map of the campus that users can explore, helping them visualize spatial layouts and navigate the area with ease. The application will be designed to be accessible even to users with minimal experience in using electronic devices.

The app will feature a search function that allows users to find specific locations quickly by entering keywords such as room numbers or department names. It will also provide basic directional guidance between selected points, streamlining the movement across campus. The system will work offline, making it convenient and reliable for users without internet access. By saving time, reducing confusion, and enhancing overall campus experience, this 3D navigation system offers a practical and cost-effective solution for improving wayfinding at Northwest Samar State University.

Objectives

The purpose of this study is to design and develop a 3D Mobile - Based Navigation system. Specifically, it aims to achieve the following:

1. To design a UI/UX – Based 3D interface.
2. To integrate a search bar feature that enables users to easily search for and locate campus areas.
3. To develop an offline navigation system.

Scope and Limitations

The scope of this project focuses on creating a 3D Navigation App for Northwest Samar State University, specifically in locating rooms, offices, and other important areas within the College of Computing and Information Science Building. However, the project has several limitations: it does not support real-time tracking and cannot display users' live locations like GPS-based apps such as Google Maps. The app is developed exclusively for Android devices and is compatible only with gaming devices that can handle 3D graphics. It is limited to English language use and confined only to specific areas of the NWSSU campus, particularly the CCIS Building. Moreover, the navigation coverage is limited to the first, second, and third floors of the building, the fourth floor is not included. Due to time and resource constraints, advanced features like voice commands and augmented reality are not included. Despite these limitations, the app is intended to provide an innovative and helpful tool for exploring essential areas and improving campus navigation.

CHAPTER 2

REVIEW OF RELATED LITERATURE/SYSTEMS

This chapter presents related literature and systems regarding the Pathfinder: Mobile-Based 3D Navigation System, which helped researchers gain sufficient knowledge for the development of the study. All information was gathered through scholarly sources, the internet, and books to develop a deeper understanding of the research problem and identify gaps, contradictions, and limitations in the literature.

Montaro et al. (2017) on VisUn-3D: user navigation with a mobile-based 3D navigation system for Northwest Samar State University (NWSSU). Both systems use 3D maps to give users a realistic and interactive way to explore places through their mobile phones. They share the same goal of helping users find locations easily and navigate unfamiliar areas using clear and user-friendly guidance. However, they differ in their scope and target users. VisUn-3D was designed for general users navigating public or city spaces, while the 3D: Mobile-Based Navigation System focuses on the university campus, especially the College of Computing and Information Technology (CCIS) building and its surroundings. The NWSSU system also includes features made for students and visitors, such as rooms, offices, and facility information, and may include an interactive guide like Directional Overlay (DORO) for a more engaging experience. Overall, both systems share the same 3D navigation concept, but the 3d mobile-based navigation system is specially adapted to meet the needs of a campus environment, making it more practical and helpful for university users.

PathFinder: Mobile-Based 3D Navigation System provides essential insights and design principles for creating an effective, user-friendly, and inclusive navigation application. The work of Shuo Ren et al. (2018) demonstrated how immersive 3D interfaces can enhance user interaction, which guided the researchers in designing an engaging 3D environment for PathFinder. Helmi et al. (2022) emphasized the importance of user-centered design, influencing PathFinder's focus on simplicity and ease of use for all types of users within the NWSSU campus. The study by Hsuan Chen et al. (2018) inspired the inclusion of accessibility considerations, ensuring that the system is inclusive and supportive of users with varying abilities. Meanwhile, Liu et al. (2021) and Nordin, Markom, and Suhaimi et al. (2021) highlighted the advantages of AR-based navigation and mobile accessibility, which helped shape PathFinder's interface to be more interactive, visually guided, and efficient for campus navigation.

Search functionality is a key feature in navigation apps, enabling users to quickly locate destinations and points of interest. The study by Rai et al. (2024), which introduced PathPilot, highlighted how a strong search feature can help users find destinations even in areas with limited Global Positioning System (GPS) access, an idea that inspired the inclusion of an effective search bar in PathFinder. Similarly, Rokhsaritalemi et al. (2020) showed that user-centered search tools improve efficiency in mixed reality campus navigation apps, guiding the researchers to design a search feature that responds quickly to user inputs. Andri et al. (2019) demonstrated how AR campus tours with built-in search bars help users locate campus landmarks faster, reinforcing the read for accessible search

tools in Pathfinder. Moreover, Haile, Desta, and Hadgu (2023) developed a mobile guide app with real-time search and route guidance, influencing Pathfinder's integration of a similar interactive search function. Lastly, De Ocampo et al. (2019) emphasized that navigation apps for Filipino users should have simple and effective search designs that match local habits, inspiring the researchers to make Pathfinder's search tool easy to use and suited for its target users at Northwest Samar State University (NWSSU). Overall, these studies guided the creation of a reliable, intuitive, and locally adapted search feature, making Pathfinder more efficient and user-centered.

Offline navigation systems are essential for campuses with limited internet connectivity or GPS availability. Alhashimi, Mahmmod, and Abdulameer (2021) developed a hybrid AR and Wi-Fi positioning system that works without continuous GPS, showing how navigation can still function effectively indoors, an idea that influenced PathFinder's offline capability design. Similarly, Lu et al. (2021) demonstrated an ARCore-powered system with offline tracking, proving that navigation can continue smoothly even under weak network conditions. The PathPilot study by Rai et al. (2024) also highlighted the need for reliable offline functions to maintain usability in GPS-limited areas, reinforcing PathFinder's goal to remain accessible at all times. Shafique and Qureshi (2022) explored using preloaded 3D maps and beacon technology for offline indoor navigation, which inspired PathFinder's use of locally stored map data to guide users without internet. Lastly, Chen et al. (2019) developed an offline campus wayfinding system focused

on efficiency and minimal resource use, supporting the researchers' aim to make PathFinder lightweight and functional even without a stable connection.

Accessibility remains a significant challenge in campus navigation tools. Chen et al. (2015) developed BlindNavi, a system made specifically for visually impaired users, showing how technology can support individuals with limited vision in navigating indoor spaces. This inspired the researchers to consider accessibility features that make PathFinder more inclusive for all users. Rokhsaritalemi et al. (2020) also highlighted the need for mixed reality navigation apps that are inclusive for differently-abled users, guiding the development of an interface that is easy to use and adaptable. Lastly, De Ocampo et al. (2019) pointed out that navigation systems in universities should address local accessibility barriers, encouraging the researchers to design PathFinder to suit Filipino users and the NWSSU campus environment.

The literature also recognizes the significance of hybrid systems in indoor environments. Alhashimi, Mahmmod, and Abdulameer (2021) developed a system that integrates Augmented Reality (AR) with Wi-Fi positioning, allowing users to navigate effectively even when GPS signals are weak or unavailable, similar to the conditions inside the CCIS building. Similarly, Haile, Desta, and Hadgu (2023) created a mobile campus guide app for Mekelle University that included real-time maps, route guidance, and multiple viewing modes, features that closely match the goals of the proposed 3D mobile-based navigation system. In addition, Lu et al. (2021) designed an ARCore-powered navigation system that provided real-time tracking and seamless indoor-outdoor transitions, delivering an immersive user

experience that the current study seeks to replicate within the NWSSU campus setting.

Recent developments in semantic 3D city modeling have improved how users understand and interact with spatial environments. Beil and Kolbe (2024) explained that semantic 3D models combine geometry, topology, and meaning to make navigation more accurate and support better decision-making. Similarly, Yao et al. (2018) presented 3D street space models that enable route optimization and context-aware guidance. Liu et al. (2022) addressed the cognitive difficulties users face in mixed reality environments and proposed design strategies to make interaction with 3D spatial data easier. Jomsri (2018) integrated augmented reality (AR) with 3D models to improve indoor navigation in university libraries, while Haile et al. (2023) developed mobile guide applications that feature real-time semantic map updates to help users understand their surroundings more effectively.

These studies align with the researchers' PathFinder: Mobile-Based 3D Navigation System, which applies semantic 3D data to create an intuitive and immersive navigation experience. By including semantic information—such as pathway types, accessibility features, and contextual landmarks—into the 3D environment, PathFinder enhances users' ability to navigate the CCIS building efficiently. The system design also follows the principles of semantic 3D modeling by focusing on accuracy, visual realism, and user-centered functionality, making it suitable for the campus environment.

Designing mobile spatial navigation systems from the user's perspective: An interdisciplinary review. Spatial Cognition & Computation. This study

emphasizes the importance of user-centered design in mobile navigation systems. Emphasized the importance of user-centered design in mobile navigation systems, influencing PathFinder's focus on simplicity and ease of use for all types of users. Ruginski, I., Giudice, N., Creem-Regehr, S., & Ishikawa, T. (2022). Design and implementation of a campus navigation application with augmented reality for smartphones. This thesis investigates the application of augmented reality in campus navigation, providing insights into UI/UX design. Investigated the application of augmented reality in campus navigation, providing insights into UI/UX design that guided the researchers in designing an engaging environment for PathFinder. Lautenschlager, B. (2012).

Campus mobile navigation system based on shortest-path algorithm and user collaborations. This focuses on implementing a shortest-path algorithm in campus navigation, improving search functionality, which inspired the inclusion of an efficient search feature in PathFinder. Len, S. N., & Zainon, W. M. N. W. (2014). Design and Implementation of a Campus Navigation System. This project explores the design and implementation of a campus navigation system with search functionality, reinforcing the importance of a well-designed search feature. Adeoyin, A. (2019).

Introduced a campus navigation system developed on the Android platform to assist students and visitors in locating buildings and facilities within a university. This study influenced PathFinder's development by demonstrating the effectiveness of mobile-based navigation applications in enhancing user convenience and accessibility. Anpat, V., Shewale, A., & Bhangale, Y. (2016).

Designed and implemented a navigational system for the University of Ilorin, focusing on guiding users efficiently within the campus environment. Their work provided insights into system design and functionality that helped shape PathFinder's goal of offering accurate and user-friendly campus navigation. Olayiwola, W. B., Jolayemi, O. A., Faruk, N., & Ayeni, J. K. (2015). In addition, development of an Android-based campus guide for Mautech, Yola, Nigeria, emphasizing accessibility and ease of navigation for students and visitors. Their findings supported PathFinder's objective of ensuring reliable and efficient navigation, even in offline. Oye, N. D., & Nathaniel, J. (2017).

In summary, previous studies shown have that 3D and AR – Based navigation system greatly improve spatial awareness, user experience, and accessibility in educational setting. However, many of these depend on advanced technology, stable internet, Global Positioning System (GPS), or costly infrastructure resources that may not always be available at College of Computing and Information Sciences (CCIS) or in other developing institutions. This research addresses that gap by creating a 3D mobile navigation app specifically designed for a resource limited text. The system prioritizes offline functionality, a lightweight structure that works on most android devices, and an inclusive interface that support both able bodied and differently abled users. By adapting proven 3D navigation technologies and design principles to local conditions, this study provides a practical accessible, and community -oriented solution tailored to the needs of Northwest Samar State University campus.

CHAPTER 3

TECHNICAL BACKGROUND

This chapter outline the software, hardware, and human resources of the 3D Navigation App. These resources are labeled as the most vital requirements for the successful creation of the application, providing its functionalities works as stipulated by the study.

Software Requirements

The researchers are expected to integrate the appropriate software tools and resources and technologies to produce the PathFinder: Mobile-based 3D Navigation app for the College of Computing and Information Sciences. The software requirements indicated herein will guarantee that the system runs accurately, provides 3D navigation, and operates seamlessly on Android devices. The software requirements, specifications, and descriptions are outlined in the table following.

Table 1
Software Requirements for the Development of 3D Mobile – Based Navigation System

Software	Specifications	Descriptions
Programming Language	JavaScript, Java, HTML, CSS	JavaScript adds interactive features to the app, html and css, are for app's structure, style, and interactive features, while Java handles core functions and Android development.

Software	Specifications	Descriptions
3D Modeling Tool	Blender	To create or edit 3D models of the CCIS building and nearby areas, and to add navigation features to help users explore the campus.
Development Environment	Capacitor Framework Visual Studio Code, Unity Editor	Capacitor is used for building and running the mobile app across platforms. Visual Studio Code is used for coding the app's functions and interface, while Unity Editor is used for creating and testing the 3D environment.

Table 1 shows the software requirements for developing the 3D Mobile–Based Navigation System for Northwest Samar State University, highlighting the key tools used to build and design the application. The programming languages include JavaScript, Java, HTML, and CSS where JavaScript provides interactive features while Java manages the core functions and Android development, while HTML and CSS define the app's structure and style to ensure a clean and user-friendly design. For creating realistic campus models, Blender is used as the 3D modeling tool, allowing developers to design and edit 3D representations of the CCIS building and surrounding areas. The development environment consists of the Capacitor Framework**,** Visual Studio Code, and Unity Editor. Capacitor enables the app to run across different platforms, Visual Studio Code is used for coding and interface development, and Unity Editor is essential for building and testing the 3D environment. Together, these tools work together to build an interactive navigation system that helps students, teachers, and visitors easily find their way around the campus.

Hardware Requirements

Hardware requirements outline the physical components required for system development and testing. Table 2 shows the hardware requirements for creating Northwest Samar State University's 3D navigation system. The hardware specifications used in system development include equipment for coding, testing, and operating the application. These devices include a laptop and a mobile device, each chosen for its ability to successfully support the system's functions. Moreover, descriptions and specifications for each hardware requirement are supplied to help readers comprehend their position in the system development process.

Table 2

Hardware Requirements for the Development of 3D Navigation System

Hardware	Specifications	Descriptions
PC	Intel Core i5 (10th Gen), 3.2 GHz, 16 GB RAM, 1TB HDD / 256 GB SSD, Windows 10/11	A desktop computer for system development, 3D modeling, and testing, offering higher performance and storage capacity.
Mobile Device	3GB RAM 64GB ROM Android 13 above	A portable device used to test the app. It is used to check how the app performs on mobile phones and ensures it works well for users.

The hardware specifications for the 3D Positioning System are essential for the development and evaluation of the application. The personal computer is primarily used for coding, 3D modeling, and testing, providing the necessary processing power and storage capacity to handle complex development tasks and

finalization of the manuscript, ensuring proper documentation of the project. Meanwhile, the mobile device serves as the testing platform, allowing the researchers to verify that the application functions effectively on smartphones. With the appropriate specifications, these devices collectively ensure that the app’s navigation and 3D features perform smoothly, are user-friendly, and meet the functional requirements of the institution.

Human Resource Requirements

Human Resource Requirements are critical to the effective development and implementation of mobile Mobile – based 3D navigation system. The team responsible for this project consists of many responsibilities each with a distinct function to guarantee that the system is accomplished as per schedule, satisfies its objectives, and functions properly. The aforementioned positions include the Project Manager, who leads and assembles the team, the Programmer, who codes and develops the app, the Designer, who concentrates on the app's appearance and user experience, the Researcher, who stays current on new trends and technologies, and the users themselves, who supply invaluable input for enhancing the app's functionality. Each individual contributes significantly to the project's achievement in general.

Table 3

Human Resource Requirements of 3D Navigation System

Human Resource	Descriptions
Project Manager	The person who leads the project, makes sure everything is on track, and helps the team stay organized. They make sure the project is completed on time and meets its goals.
Programmer	In charge of the actual coding and development of the mobile app. The programmer will work with 3D modeling tools, the Unity engine, and other technologies to create the navigation system and ensure its functionality.
Designer	The person who creates how the app looks. They make sure the app is easy to use, good to look at, and organized well for user.
Researcher	Conducts research on the latest trends and technologies related to navigation systems, 3D modeling, and campus apps. They will gather information, review literature, and provide insights to improve the design and functionality of the app.
User	The people who will use the app, like students, teachers, and visitors. They give feedback on how easy or helpful the app is, which helps make the app better.

The human resource requirements for the 3D Navigation System are vital for the prompt accomplishment of the project. Every individual share it's fair amount of roles and responsibilities that rectify the seamless development and effectiveness of the application, cementing its imminent success. The project manager is responsible for leading the team, organizing tasks, and ensuring that the project stays on the schedule and meets its objectives. This person oversees that all goals and deadlines are achieved efficiently. The programmer handles the coding and technical development of the mobile application. This includes integrating 3D models, building the navigation features, and ensuring that system

functions smoothly using tools such as blender, visual studio, and other development technologies.

Gantt chart

The researchers developed a project timeline represented by a Gantt chart. This chart outlines the phases and tasks necessary for achieving the successful development and optimal performance of the proposed system.

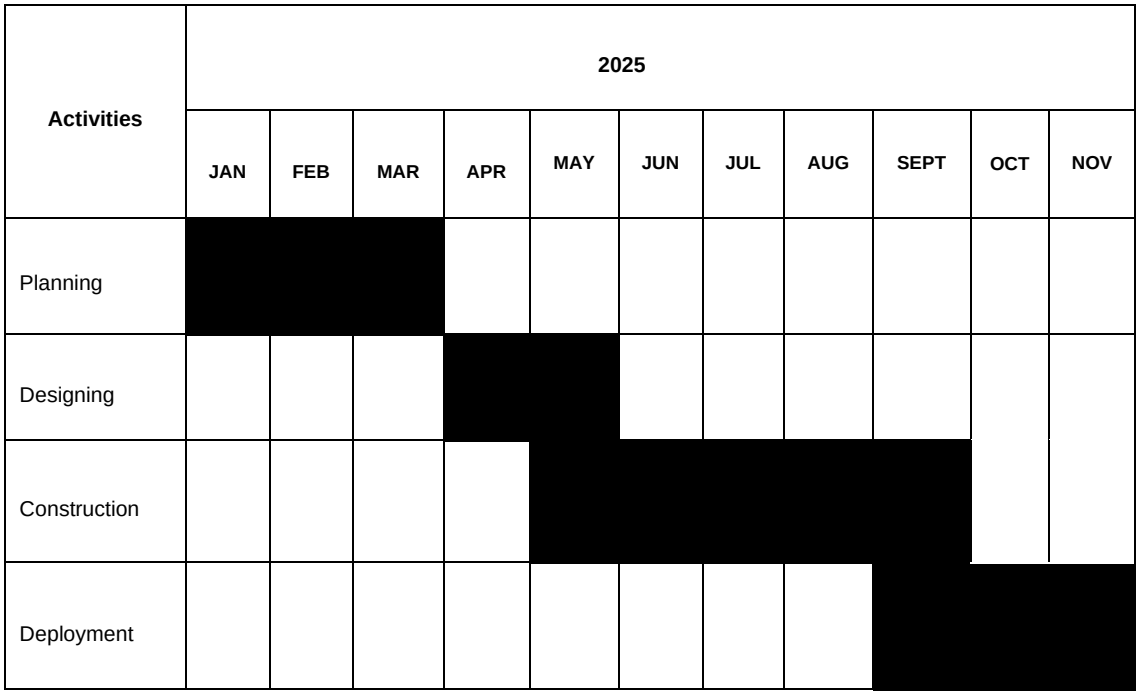


Figure 1. Gantt Chart for the Development and Implementation of Mobile-Based 3D Navigation System

The Gantt Chart in Figure 1 illustrates the development schedule for the PathFinder: Mobile-Based 3D Navigation in College of Computing and information Sciences at Northwest Samar State university, covering the months from January to November 2025. The project is divided into four main phases: Planning, Designing, Construction, and Deployment. Planning occurred during January and

March focusing on identifying user needs, defining system goals, outlining the app’s key features, and preparing for proposal defense. The Designing phase, which spanned from April to May, involved creating the user interface, developing the 3D layout of the College of Computing and Information Sciences (CCIS) building, and finalizing the application's structure. This phase overlapped with Planning, following the Rapid Application Development (RAD) approach, which encourages iterative and parallel progress.

From May to September the Construction phase took place, where developers focused on coding, integrating 3D models in blender, and implementing core features such as search and navigation. This phase also included testing and refinement through multiple iterations. Finally, the Deployment phase was scheduled for September and October, which involved final testing, debugging, documentation, and release readiness. The Gantt Chart provided a clear and organized timeline that guided the development process, ensured timely completion of project milestones, and supported continuous collaboration and feedback among team members.

CHAPTER 4

METHODOLOGY, RESULTS, AND DISCUSSION

In developing the PathdFinder: Mobile: Based 3D Navigation System, the researchers utilized development, adaptable requirements, and prototyping to meet all the project objectives. The Rapid Application Development (RAD) methodology guided the process, enabling the team collect requirements concurrently with app development. This approach allowed the researchers to integrate feedback and make adjustment throughout the project.

The RAD model emphasizes prototyping, user involvement, and quick feedback, making it suitable for developing a system that prioritizes usability and accessibility. This approach ensures that the application can be continuously refined based on user input while maintaining efficiency in its design and implementation

Requirements Analysis

In this phase, the primary functions that the system must perform to achieve the development objectives were clearly defined. The researchers carefully analyzed the requirements of the PathFinder: Mobile-Based 3D Navigation System based on the feedback and recommendations provided by the project panel, as well as the observed challenges in navigating the College of Computing and Information Sciences building. The panel suggested developing a system that would help students, faculty, and visitors locate rooms, offices, and facilities more

efficiently. Observations of the current navigation process revealed that users often rely on printed maps, static signage, or verbal directions, which can lead to confusion and delays. These traditional tools also lack interactivity and real-time guidance.

Based on the panel's recommendations and the identified navigation challenges, the researchers determined and categorized both user and system requirements in line with the objectives of the study. The analysis focused on identifying the essential features, performance, and usability requirements that would make the system effective and reliable. By addressing these requirements, the proposed 3D Navigation System ensures that the application aligns with the needs of its users. It guarantees that the system will not only help users navigate the CCIS building with ease but also provide an efficient, accessible, and user-friendly alternative to traditional navigation tools.

Requirements Documentation

Requirement documentation outlines both the functional and non- functional requirements that guide the development process and ensures the projects meets its objectives. The functional requirements describe the core tasks of the system, such as searching for rooms, navigating between points, and displaying an interactive 3D map. Meanwhile, the non-functional requirements focus on the system's quality attributes, including usability, performance, reliability, and security. Clearly defining these requirements helps the development team stay

focused, avoid misunderstandings, and build a system that is useful, efficient, and easy to use.

Functional Requirements

The functional requirements define the core features and capabilities that the 3D Mobile-Based Navigation System must provide to meet the needs of its users. These requirements focus on the essential operations that enable students, faculty, and visitors to effectively navigate the CCIS building. They describe the main functions that the system should perform, such as displaying a 3D map, allowing users to search for locations, and providing directional guidance.

Table 4

Functional Requirements for the Mobile-Based 3D Navigation System

Function	Descriptions
Search Location Bar	Allows users to quickly search for specific locations (e.g., room numbers, offices, departments) using keywords.
Offline Navigation	To navigate pathway even without an active internet connection.

The table 4 shows the functional requirements for the Mobile-Based 3D Navigation System. The search bar feature allows users to locate specific location within the college of computing and information sciences (CCIS), such as department, rooms, offices, and laboratories. It provides faster access to locate the destinations. The offline navigation features enable users to navigate pathway

even without an active internet connection. This function ensures the usability, especially in areas with poor or no network connection, users can still navigate the campus efficiently without depending on mobile data or Wi - Fi.

Non-Functional Requirements

The non-functional requirements describe the quality attributes of the 3D Navigation System. Unlike functional requirements that specify what the system should do, non-functional requirements define how well the system performs under different conditions. These include performance, security, reliability, usability, and compatibility to ensure the application is effective, secure, and accessible to a wide range of users.

Table 5

Non- Functional Requirements for the Mobile-Based 3D Navigation System

Non-Function	Descriptions
Performance	The system should operate smoothly on Android devices without significant delays, even when rendering 3D models.
Reliability	The app should function consistently offline, ensuring access without internet connectivity.
Usability	The interface must be simple and intuitive so that even first-time users can easily navigate the app.

The table 5 presents the non-functional requirements for the 3D Navigation System, which emphasize performance, security, reliability, usability, and compatibility to ensure its effectiveness and accessibility. The performance requirements ensures that the system should run smoothly on android devices, rendering 3D models without lag to ensure a responsive user experience. The reliability must the application work consistency even offline, providing uninterrupted navigation access without internet connection. The usability interface should be simple and intuitive, allowing first time users to easily navigate and find their destinations.

Design of Software, System, Product, and/or Processes

This phase presented the design of the Mobile-Based Navigation System, focusing on the development of the software, system architecture, and core features. The system serves as a conceptual model that outlines the application's structure, behavior, and essential characteristics, acting as a detailed blueprint for the implementation process. During this design phase, the researchers defined the system components, data flow, and interactions, ensuring that the application aligns with both functional and non-functional requirements.

System Architecture

This is where researchers brainstormed how to build the system and the operate it. The researchers decide what design they would apply to the application regarding software, hardware, function and interface requirements. The system architecture was used for the researchers to have an overview of the system.

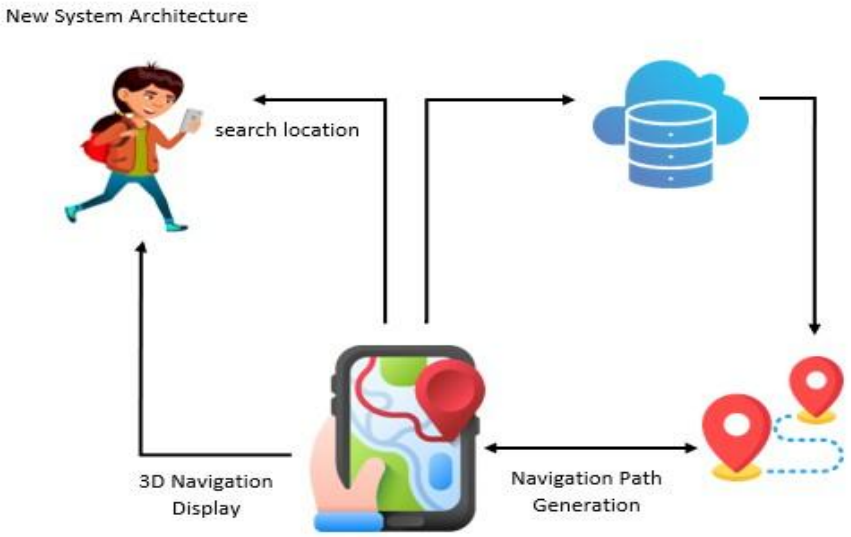


Figure 2. System Architecture of Pathfinder: Mobile-Based 3D Navigation System

Figure 2 shows how the Pathfinder app is set up. The diagram shows the users using the mobile app to search for a room or place from Point A to Point B. The app sends the request to the 3D Map Storage and the Route Finder, which work together to get the room details and find the best path. The information is then sent back to the app so the user can see an interactive 3D map with the route on their Android phone. This setup helps students, visitors, and staff easily find rooms and offices inside the CCIS building even without internet.

Process Model

The system model represents or shows the system's flow, wherein the module is related to its environment. To understand the system, researchers created different charts. These charts include context diagram, data flow, and

functional decomposition diagram. The figures display the activities and represent the flow model during the actual system development.

Context Diagram

Figure 3 shows the Context Diagram of PathFinder: Mobile – Based 3D Navigation System. This diagram presents the overall structure of the system, illustrating how users interact with the application, it provides a simplified view of the entire process showing the input and output involved n the navigation features.

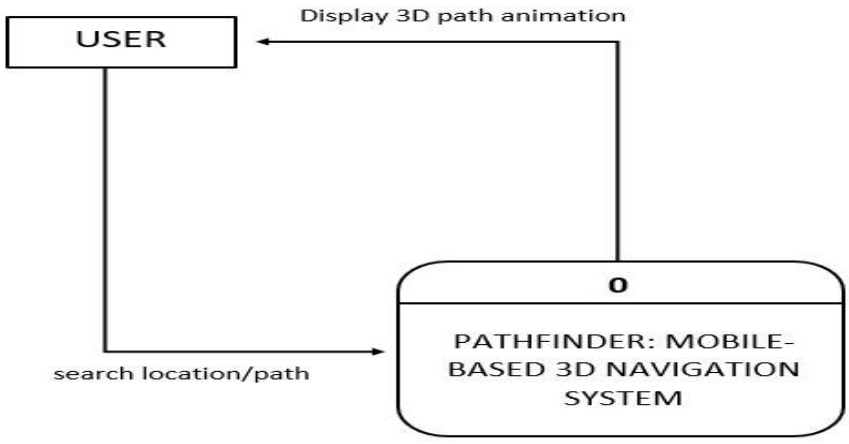


Figure 3. Context Diagram of Pathfinder: Mobile-Based 3D Navigation System

In context diagram, the user is the primary external entity that interacts with the pathfinder system. The user searches or locates starting point and destination within the application. This input data is sent to the PathFinder: Mobile – Based 3D Navigation System, which processes the request to determine the appropriate route. After processing, the system responds by displaying the pathway animation that visually guides the user from point A to point B.

Data Flow Diagram

The Data Flow Diagram (DFD) for the PathFinder: Mobile - Based 3D Navigation System illustrates how data moves throughout the system and how users interact with it. It shows the main processes, data stores, and information flow between each component. The DFD serves as a visual representation of how user inputs are processed and how the system retrieves, analyzes, and outputs navigation data to guide users within the campus.

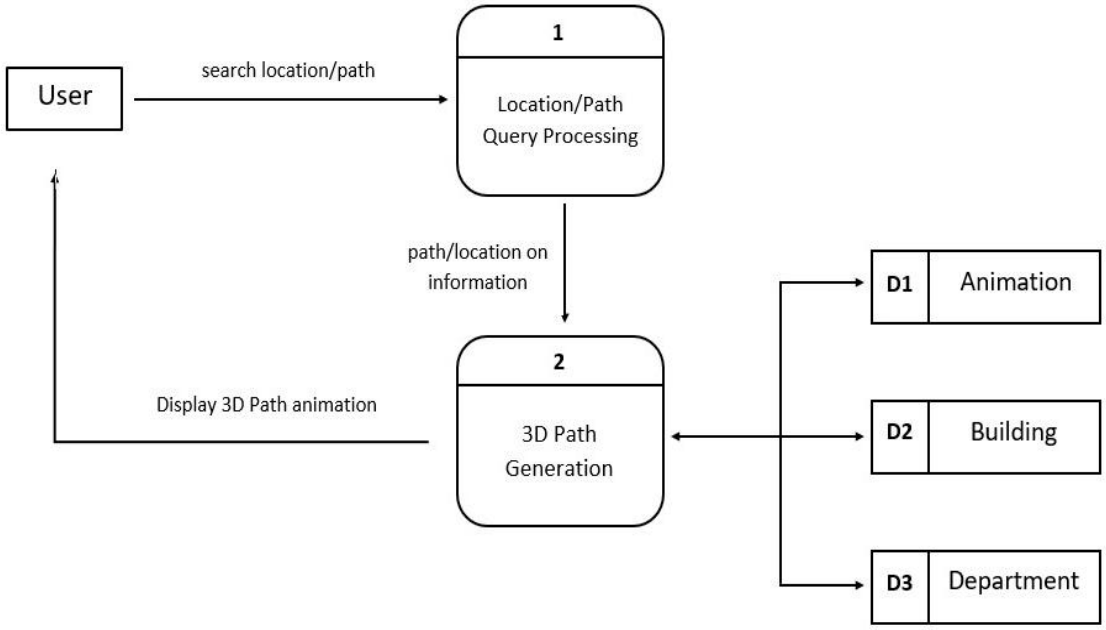


Figure 4. Data Flow Diagram of Pathfinder: Mobile-Based 3D Navigation System

Figure 4 shows the Data Flow Diagram of the Pathfinder: Mobile-Based 3D Navigation System. This diagram shows how the PathFinder: Mobile-Based 3D Navigation System works by showing the flow of data between the user and the system. First, the user enters a location or selects two points (Point A to Point B). The system then generates a map or path using the data stored in its databases. It gathers path details from three data sources. D1 (Department) – contains

information about rooms, offices, and departments. D2 (Building) – stores the 3D map data of the building. D3 (Animation) – provides the visual animation used to display the 3D path on the screen. After collecting the needed information, the system shows the path and navigation details back to the user. This process helps users easily find their destinations within the CCIS building even without real-time GPS.

Functional Decomposition Diagram

The Functional Flow Diagram (FFD) for the PathFinder: Mobile-Based 3D Navigation System shows how users interact with the system to search for locations and navigate within the NWSSU-CCIS building. It highlights key features such as Search**,** Point A to Point B Navigation, and Offline Navigation, presenting the overall flow of activities from user input to navigation output.

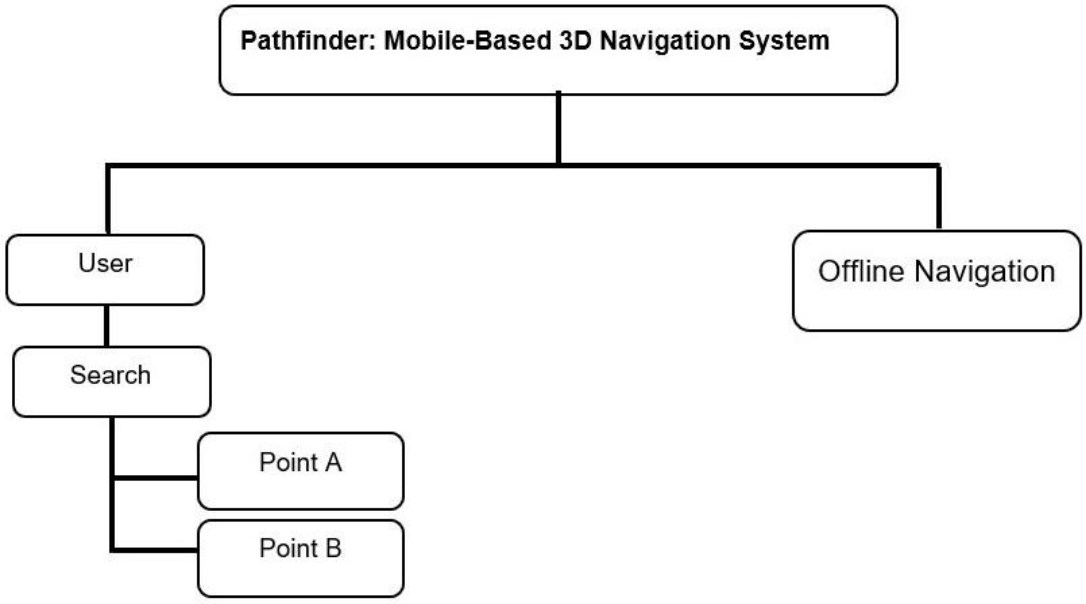


Figure 5. Functional Decomposition Diagram of Pathfinder: Mobile-Based 3D Navigation System

Figure 5 shows the Functional Decomposition Diagram of the Pathfinder: Mobile-Based 3D Navigation System. It contains all the functions and tasks within the app. the process begins when the User searches for a location by entering a start (Point A) and destination (Point B). The system then generates a 3D navigation route to guide the user. Even without an internet connection, the Offline Navigation feature allows users to access preloaded maps and directions. Finally, the app displays a 3D path and step-by-step guide, ensuring smooth and efficient navigation around the campus.

Data Model

The data model of the Pathfinder: Mobile-Based 3D Navigation System defines how information is organized and connected inside the app. It includes key entities such as Room (stores details like Room ID, room name, building, and floor), Route **(**contains the path between two rooms including distance and path details), and UserRequest (records each search request with the starting and destination rooms). The UserRequest links to two rooms as the starting point and destination, while the Route connects one start room to one end room. This structure allows the system to process user searches, calculate routes, and display accurate 3D maps and directions.

Entity Relationship Diagram

An Entity Relationship Diagram (ERD) was a graphical representation of entities and their relationships to each other, used in database design. It was a tool for modeling the relationships between data entities in a database. An ERD

typically consisted of entities, which represented real-world objects or concepts, and the relationships between them.

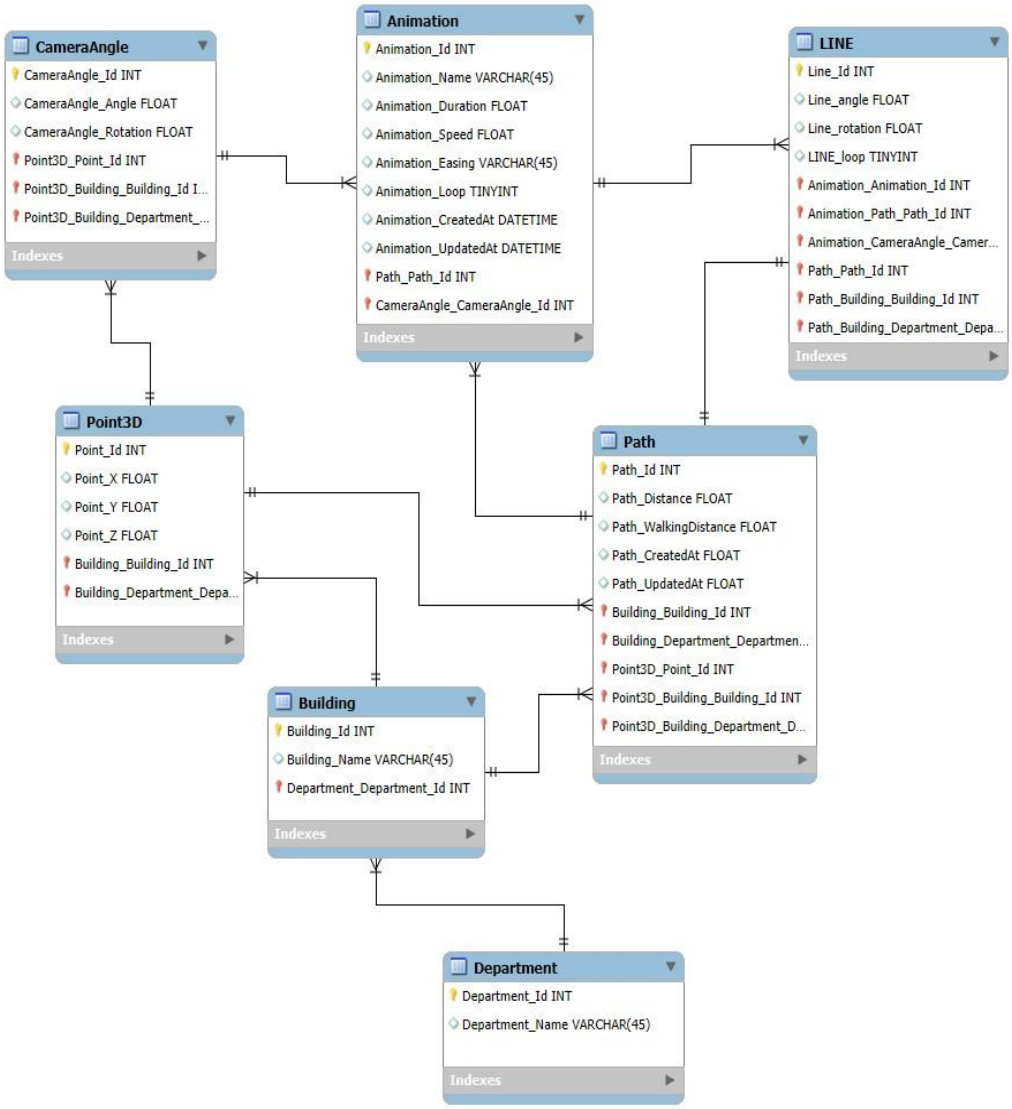


Figure 6. Entity Relationship of Pathfinder: Mobile-Based 3D Navigation System

Figure 6 presents the entity–relationship diagram of the PathFinder. A building can contain several departments, while each department belongs to a single building. Every building is mapped to a specific 3D point that defines its exact location on the campus map. A path is composed of multiple lines that connect starting and ending 3D points, and each line is drawn between two 3D coordinates. An animation is linked to a camera angle, a path, and a line to control how the camera moves and rotates within the 3D environment. Camera angles specify the view settings for a particular 3D point, allowing users to experience smooth navigation along the defined paths.

Data Dictionary

The Data Dictionary is a centralized document that gives detailed information about the structure, organization, and attributes of the data used in the Pathfinder: Mobile-Based 3D Navigation System. It includes clear definitions and descriptions of all the data inside the app, such as tables, fields, and data types. This dictionary is very important for data management because it helps everyone understand the data, maintain its quality, and communicate effectively among developers, researchers, and other stakeholders.

Table 6 show the data dictionary for Animation table of Pathfinder: Mobile-Based 3D Navigation System.

Table 6

Data Dictionary of Animation Table of Pathfinder: Mobile-Based 3D Navigation System

Field Name	Data Type	Null	Description	Key	Reference Table
Animation	Int	No	Unique identifier for animation.	PK	—
Animation_Name	Varchar	No	Name of the animation (e.g., camera mov).		—
Duration	Float	Yes	Duration of the animation in second.		—
Speed	Float	Yes	Movement speed of the animation.		—
Easing	Varchar	Yes	Easy type (e.g., linear, ease-in).		—
Loop	Boolean	Yes	Indicates if the animation repeats.		—
CreatedAt	Datetime	Yes	Date and time when the animation was created		—
UpdatedAt	Datetime	Yes	Date and time when the animation was last updated.		—

Table 6 – The Animation table stores all the movement effects used in the 3D navigation system. It shows the name of the animation, its duration (how long it runs), speed (how fast it plays), easing type (how it starts and stops smoothly),

and whether it will loop continuously. It also records the date created and date updated to track changes. Each record in this table represents one specific animation that can be applied to paths, camera angles, or objects to create smooth transitions inside the 3D map.

Table 7 show the data dictionary for CameraAngle table of Pathfinder: Mobile-Based 3D Navigation System.

Table 7

Data Dictionary of CameraAngle Table of Pathfinder: Mobile-Based 3D Navigation System

Field Name	Data Type	Null	Description	Key	Reference Table
CameraAngle_Id	Int	No	Unique identifier for each camera angle.	PK	—
CameraAngle_Angle	Float	Yes	Angle of the camera in degrees.		—
CameraAngle_Rotation	Float	Yes	Rotation value for camera orientation.		—

Table 7 – The CameraAngle table stores the different camera views used in the 3D navigation system. It includes the camera name**,** angle**,** rotation, and the date created/updated to track modifications. Each record represents a specific camera position that controls how users see the 3D environment, allowing smooth viewing of buildings, paths, and navigation routes from various perspectives.

Table 8 show the data dictionary for Path table of Pathfinder: Mobile-Based 3D Navigation System.

Table 8

Data Dictionary of Path Table of Pathfinder: Mobile-Based 3D Navigation System

Field Name	Data Type	Null	Description	Key	Reference Table
Path_Id	Int	No	Unique identifier for path.	PK	—
Path_Distance	Float	Yes	Total length of the path		—
Path_WalkingDistance	Float	Yes	Walking distance for navigation.		—
Path_CreatedAt	Datetime	Yes	Date and time when the path was created.		—
Path_ UpdatedAt	Datetime	Yes	Date and time when the path was updated.		—

Table 8 – The Path table stores the navigation routes within the 3D campus map. It includes details such as the path name**,** distance**,** walking distance, and the date created/updated to track changes. Each record represents a specific route that connects starting and ending points, helping the system guide users along the correct path inside the 3D environment.

Table 9 show the data dictionary for Point3D table of Pathfinder: Mobile-Based 3D Navigation System.

Table 9

Data Dictionary of Point3D Table of Pathfinder: Mobile-Based 3D Navigation System

Field Name	Data Type	Null	Description	Key	Reference Table
Point_Id	Int	No	Unique identifier for each 3d point.	PK	—
Point_x	Float	No	X-coordinate space.		—
Point_y	Float	No	X-coordinate space.		—
Point_Z	Floate	No	X-coordinate space.		—

Table 9 – the Point3D table stores the exact 3D coordinates used to map locations in the virtual campus. It includes the point name**,** x**,** y, and z values, along with the date created/updated to track changes. Each record represents a specific point in the 3D environment, serving as a reference for creating paths, positioning buildings, or setting camera angles within the navigation system.

Table 10 show the data dictionary for Line table of Pathfinder: Mobile-Based 3D Navigation System.

Table 10
Data Dictionary of Line Table of Pathfinder: Mobile-Based 3D Navigation System

Field Name	Data Type	Null	Description	Key	Reference Table
Line_ id	Int	No	Unique identifier for each line segment.	PK	—
Line_ Angle	Float	Yes	Angle between connected point.		—
Line_ Rotation	Float	Yes	Rotation value along the path.		—

Table 10 – The Line table stores the line segments that connect different 3D points to form paths in the navigation system. It includes the line name, the starting point (Point3D), the ending point (Point3D)**,** angle**,** rotation**,** and the date created/updated. Each record represents a specific connection between two 3D points, helping define the exact direction and structure of routes within the 3D environment.

Table 11 show the data dictionary for Building table of Pathfinder: Mobile-Based 3D Navigation System.

Table 11
Data Dictionary of Building Table of Pathfinder: Mobile-Based 3D Navigation System

Field Name	Data Type	Null	Description	Key	Reference Table
Building_ Id	Int	No	Unique identifier for each camera angle.	PK	—
building_ Name	Float	Yes	Angle of the camera in degrees.		—

Table 11 – The Building table stores information about all the buildings included in the 3D navigation system. It contains the building name, a unique building ID, and the date created/updated to track any changes. Each record represents one specific building on the campus, serving as a main reference for linking departments, rooms, and navigation paths within the 3D environment.

Table 12 show the data dictionary for Building table of Pathfinder: Mobile-Based 3D Navigation System.

Table 12

Data Dictionary of Department Table of Pathfinder: Mobile-Based 3D Navigation System

Field Name	Data Type	Null	Description	Key	Reference Table
Department _ Id	Int	No	Unique identifier for each department.	PK	—
Department _ Name	Varchar	No	Name of the department		—

Table 12 – The Department table stores the list of departments located within each building in the 3D navigation system. It includes the department name, the building ID it belongs to, and the date created/updated to track modifications. Each record represents a specific department and links it to its corresponding building, helping users identify the exact location of departments within the campus during navigation.

Implementation Results

This chapter presents the implementation process and results of the developed Pathfinder: Mobile-Based 3D Navigation System for the College of Computing and Information Sciences at Northwest Samar State University. It discusses how the proposed system was developed based on the identified requirements and objectives of the study. The implementation phase includes the integration of system components such as 3D modeling, interface design, and navigation functions using appropriate tools and technologies. Furthermore, this chapter highlights the outcomes of the system's functionality, performance, and usability testing. The results demonstrate how the system effectively assists users in locating rooms, offices, and other key areas within the CCIS building, providing an innovative and efficient alternative to traditional navigation methods.

The developed system was implemented, and all three objectives of the study were met. The result of implementing the system are as follows:

1. To design UI/UX – Based 3D interface

This objective aims to create a 3D interface that looks good and is easy to use. The goal is to make users feel comfortable while exploring the campus in a virtual way. A well-designed UI and UX will help users find what they need quickly and make navigation more enjoyable.

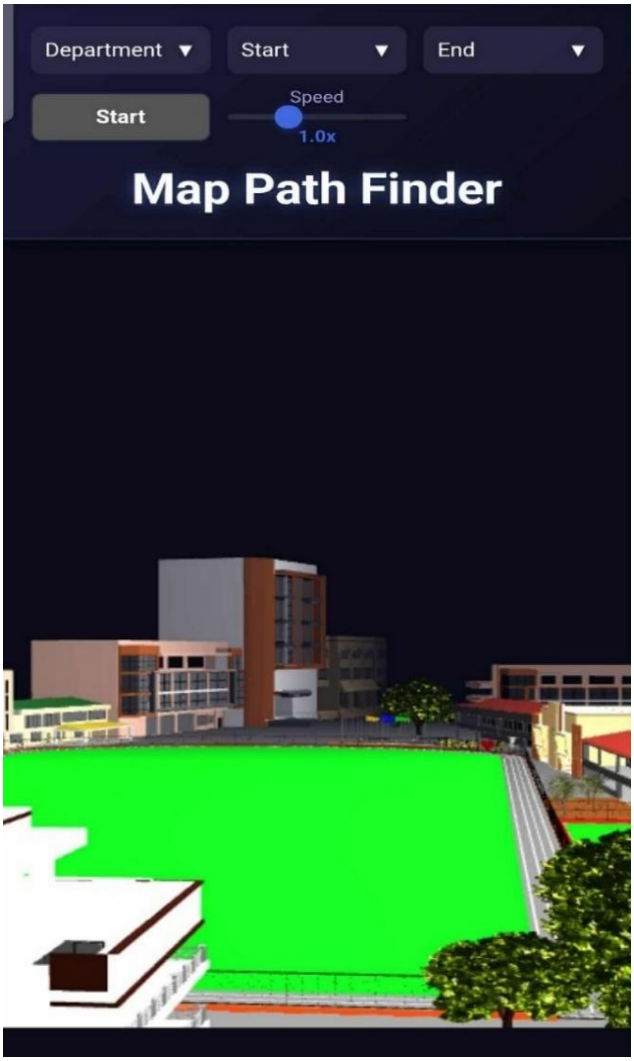


Figure 7. UI/ UX – Based 3D interface for Mobile App Design

Figure 7 shows the result of the first objective of the app, which is to create a UI/UX-based 3D interface. This means the application was designed with a clear

and easy-to-use layout (UI) and a smooth, user-friendly experience (UX) while displaying a 3D map of the campus.

The developed system was evaluated at Northwest Samar State University to determine its level of usability and effectiveness among its intended users. The evaluation involved a total 330 respondents, composed of 232 for digital form and 98 for paper- based survey.

To ensure clarity and consistency, the researchers provided a guided online demonstration through a digital form and paper-based survey. This form presented a detailed overview of the system's features, including selecting department, rooms from point A and point B. After reviewing the workflow and interface, the respondents answered the System Usability Scale (SUS) questionnaire. The instrument contained ten statements rated on a five-point Likert scale ranging from Strongly Disagree to Strongly Agree.

Table 13

Summary of System Usability Scale for PathFinder Mobile Application

Statement	SD	D	U	A	SA
1. I think that I would like to use this website frequently.	0	5	33	90	202
2. I found this website unnecessarily complex.	100	126	64	30	10
3. I thought this system was easy to use.	3	10	22	150	145
4. I think that I would need assistance to be able to use this website.	140	100	40	30	20

5. I found the various functions in this website were well integrated.	3	17	32	22	156
6. I thought there was too much inconsistency in this website.	120	120	60	30	0
7. I would imagine that most people would learn to use this website very quickly.	1	16	20	136	157
8. I found this system very cumbersome/awkward to use.	120	120	42	27	21
9. I felt very confident using this website.	0	16	28	123	163
10. I needed to learn a lot of things before I could get going with this website.	145	106	40	25	14

Where:
SD-Strongly Agree; D-Disagree; U-Undecided; A - Agree; SA-Strongly Agree

N = 330; **Overall Average SUS Score:** 78.07

Table 16 presents the evaluation results from the Pathfinder: Mobile- Based 3D Navigation System. The group obtained an overall SUS score of 79.08, which exceeds the benchmark value of 70. This indicates that the system may be considered to have good to excellent usability. Respondent generally agreed that the system's interface was clear, intuitive, and well-organized, allowing users to navigate the CCIS building efficiently. While a few minor complexities were reported, these did not significantly impact the overall usability of the system.

Overall, the evaluation results demonstrate that Pathfinder effectively achieved its intended purpose helping students, faculty, and visitors locate rooms, offices,

and other important areas more easily, reducing confusion, and improving efficiency in campus navigation. The high SUS score confirms that the system is user-friendly, easy to learn, and practical for everyday use within the NWSSU environment.

2. To integrate a search bar feature that enables users to easily search for and locate campus areas.

This objective focuses on adding a search bar to the system. It will allow users to type the name of a room, office, or building and find it right away. This feature will make navigation faster and help users save time while exploring the campus.

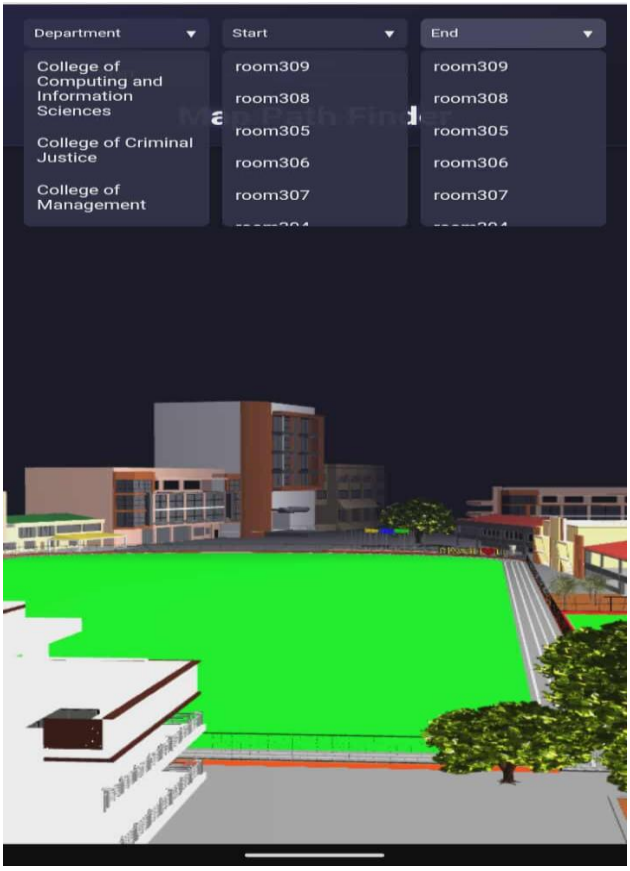


Figure 8. Search Feature of Mobile App

Figure 8 shows the result of the second objective of the app, which is to add a search feature**.** This feature lets users select from a dropdown list of campus areas, making it easier to quickly find and choose the correct room or location. It helps students locate places faster and view the 3D directions without searching the whole map manually.

3. To develop an offline navigation system.

This objective aims to make the app work even without an internet connection. It will help users access maps and directions anytime, especially in places with weak or no Wi-Fi. This ensures that navigation remains smooth and reliable wherever they are on campus.



Figure 9. Offline Navigation of Mobile App

Figure 9 shows the result of the second objective of the app, which is to create an offline navigation system. This feature allows users to search for locations and view 3D directions even without an internet connection. It ensures that students can still find rooms or buildings on campus anytime, making the app reliable and useful in areas with weak or no Wi-Fi.

CHAPTER 5
RECOMMENDATIONS

After conducting the study and developing the Pathfinder: Mobile-Based 3D Navigation System, the researchers focused on the main goals of the project and found that the app can help students and visitors overcome challenges in finding rooms, offices, and facilities inside NWSSU. The system is easy to use and successfully guides users to their destinations. However, the mobile application still has some minor issues, so the researchers suggest several improvements for future development.

The researchers recommend expanding the Pathfinder app so that it will cover the entire campus, not only the CCIS building. This will help more students and visitors easily locate different departments, offices, and facilities within the university. They also suggest adding voice-assisted navigation to provide spoken directions for users, especially those who have difficulty reading the screen or those who are moving while exploring the campus. These improvements will make the app more helpful, convenient, and accessible for everyone in Northwest Samar State University.

To ensure reliability and performance, the researchers recommend optimizing the application to handle larger data loads, minimize bugs, and improve responsiveness when zooming, rotating, or navigating in real time across various mobile devices. Implementing these enhancements will make the Pathfinder app more efficient, visually appealing, and user-friendly, thereby maximizing its usefulness for the university community.

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APPENDICES

A. Relevant Source Code

A. Path Selection Interface

```
<!DOCTYPE html>
<html lang="en">
<head>
   <meta charset="UTF-8">
      <meta name="viewport" content="width=device-width, initial-
scale=1.0">
   <title>3D Path Finder</title>
   <link rel="stylesheet" href="style.css">
</head>
<body>
   <div id="app">
      <div id="controls">
         <h2>3D Path Finder</h2>
         <div class="control-group">
            <label for="startPoint">Start Point:</label>
            <select id="startPoint">
               <option value="gate">Gate Entrance</option>
               <option value="faculty">Faculty</option>
               <option value="dean">Dean's Office</option>
               <option value="conference">Conference</option>
               <option value="parking">Parking Lot</option>
               <option value="lab1">Laboratory 1</option>
               <option value="lab2">Laboratory 2</option>
               <option value="speech">Speech Laboratory</option>
               <option value="lab3">Laboratory 3</option>
                              <option value="multimedia">Multimedia
Laboratory</option>
                        <option value="multimediaRoom">Multimedia
Room</option>
               <option value="301">Room 301</option>
               <option value="302">Room 302</option>
               <option value="303">Room 303</option>
               <option value="304">Room 304</option>
               <option value="csc">CSC Office</option>
               <option value="305">Room 305</option>
               <option value="306">Room 306</option>
               <option value="307">Room 307</option>
               <option value="308">Room 308</option>
               <option value="309">Room 309</option>
               <option value="401">Room 401</option>
               <option value="402">Room 402</option>
               <option value="403">Room 403</option>
               <option value="404">Room 404</option>
            </select>
         </div>
         <div class="control-group">
            <label for="endPoint">End Point:</label>
            <select id="endPoint">
               <option value="gate">Gate Entrance</option>
               <option value="faculty">Faculty</option>
               <option value="dean">Dean's Office</option>
               <option value="conference">Conference</option>
```

```
                <option value="parking">Parking Lot</option>
                <option value="lab1">Laboratory 1</option>
                <option value="lab2">Laboratory 2</option>
                <option value="speech">Speech Laboratory</option>
                <option value="lab3">Laboratory 3</option>
                                <option    value="multimedia">Multimedia
Laboratory</option>
                            <option   value="multimediaRoom">Multimedia
Room</option>
                <option value="301">Room 301</option>
                <option value="302">Room 302</option>
                <option value="303">Room 303</option>
                <option value="304">Room 304</option>
                <option value="csc">CSC Office</option>
                <option value="305">Room 305</option>
                <option value="306">Room 306</option>
                <option value="307">Room 307</option>
                <option value="308">Room 308</option>
                <option value="309">Room 309</option>
                <option value="401">Room 401</option>
                <option value="402">Room 402</option>
                <option value="403">Room 403</option>
                <option value="404">Room 404</option>
            </select>
        </div>
        <button id="findPath">Find Path</button>
        <div id="pathInfo">
            <p>Distance: <span id="distance">-</span> meters</p>
            <p>Estimated Time: <span id="time">-</span> minutes</p>
        </div>
    </div>

    <div id="viewer">
        <div id="floorSelector">
                    <button   class="floor-btn   active"   data-floor="1">1st
Floor</button>
            <button class="floor-btn" data-floor="2">2nd Floor</button>
            <button class="floor-btn" data-floor="3">3rd Floor</button>
        </div>
        <canvas id="buildingCanvas"></canvas>
    </div>
  </div>

  <script src="script.js"></script>
</body>
</html>
```

B. User Interface Layout

```
* {
  margin: 0;
  padding: 0;
  box-sizing: border-box;
}
body {
  font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
  background-color: #f5f5f5;
  color: #333;
```

```
    overflow: hidden;
}
#app {
    display: flex;
    height: 100vh;
}
#controls {
    width: 300px;
    padding: 20px;
    background-color: #fff;
    box-shadow: 2px 0 5px rgba(0,0,0,0.1);
    z-index: 10;
    overflow-y: auto;
}
#controls h2 {
    margin-bottom: 20px;
    color: #2c3e50;
    text-align: center;
}
.control-group {
    margin-bottom: 15px;
}
.control-group label {
    display: block;
    margin-bottom: 5px;
    font-weight: 500;
}
.control-group select {
    width: 100%;
    padding: 8px;
    border: 1px solid #ddd;
    border-radius: 4px;
    font-size: 14px;
}
#findPath {
    width: 100%;
    padding: 10px;
    background-color: #3498db;
    color: white;
    border: none;
    border-radius: 4px;
    font-size: 16px;
    cursor: pointer;
    transition: background-color 0.3s;
}
#findPath:hover {
    background-color: #2980b9;
}
#pathInfo {
    margin-top: 20px;
    padding: 15px;
    background-color: #f8f9fa;
    border-radius: 4px;
}

#pathInfo p {
    margin-bottom: 8px;
}
```

```
#viewer {
    flex: 1;
    position: relative;
    background-color: #ecf0f1;
}
#floorSelector {
    position: absolute;
    top: 20px;
    left: 20px;
    z-index: 5;
    display: flex;
    gap: 5px;
}
.floor-btn {
    padding: 8px 15px;
    background-color: rgba(255, 255, 255, 0.8);
    border: 1px solid #ddd;
    border-radius: 4px;
    cursor: pointer;
    transition: all 0.3s;
}
.floor-btn.active {
    background-color: #3498db;
    color: white;
}
#buildingCanvas {
    width: 100%;
    height: 100%;
    display: block;
}
```

C. Pathfinding Logic

```
const buildingData = {
    points: {
        gate: { name: "Gate Entrance", x: 10, y: 50, floor: 1 },
        faculty: { name: "Faculty", x: 30, y: 30, floor: 1 },
        dean: { name: "Dean's Office", x: 50, y: 30, floor: 1 },
        conference: { name: "Conference", x: 70, y: 30, floor: 1 },
        parking: { name: "Parking Lot", x: 90, y: 50, floor: 1 },
        lab1: { name: "Laboratory 1", x: 30, y: 70, floor: 1 },
        lab2: { name: "Laboratory 2", x: 50, y: 70, floor: 1 },
        speech: { name: "Speech Laboratory", x: 70, y: 70, floor: 1 },
        lab3: { name: "Laboratory 3", x: 30, y: 90, floor: 1 },
         multimedia: { name: "Multimedia Laboratory", x: 50, y: 90, floor: 1
},
        multimediaRoom: { name: "Multimedia Room", x: 70, y: 90, floor: 1
},
        301: { name: "Room 301", x: 30, y: 30, floor: 2 },
        302: { name: "Room 302", x: 50, y: 30, floor: 2 },
        303: { name: "Room 303", x: 70, y: 30, floor: 2 },
        304: { name: "Room 304", x: 90, y: 30, floor: 2 },
        csc: { name: "CSC Office", x: 30, y: 50, floor: 2 },
        305: { name: "Room 305", x: 50, y: 50, floor: 2 },
        306: { name: "Room 306", x: 70, y: 50, floor: 2 },
        307: { name: "Room 307", x: 90, y: 50, floor: 2 },
        308: { name: "Room 308", x: 30, y: 70, floor: 2 },
        309: { name: "Room 309", x: 50, y: 70, floor: 2 },
```

```
            401: { name: "Room 401", x: 30, y: 30, floor: 3 },
            402: { name: "Room 402", x: 50, y: 30, floor: 3 },
            403: { name: "Room 403", x: 70, y: 30, floor: 3 },
            404: { name: "Room 404", x: 90, y: 30, floor: 3 }
        },
        connections: [
            { from: "gate", to: "faculty" },
            { from: "faculty", to: "dean" },
            { from: "dean", to: "conference" },
            { from: "conference", to: "parking" },
            { from: "faculty", to: "lab1" },
            { from: "dean", to: "lab2" },
            { from: "conference", to: "speech" },
            { from: "lab1", to: "lab3" },
            { from: "lab2", to: "multimedia" },
            { from: "speech", to: "multimediaRoom" },

            { from: "gate", to: "301", stairs: true },
            { from: "parking", to: "304", stairs: true },
            { from: "301", to: "401", stairs: true },
            { from: "304", to: "404", stairs: true },

            { from: "301", to: "302" },
            { from: "302", to: "303" },
            { from: "303", to: "304" },
            { from: "301", to: "csc" },
            { from: "302", to: "305" },
            { from: "303", to: "306" },
            { from: "304", to: "307" },
            { from: "csc", to: "308" },
            { from: "305", to: "309" },

            // Third floor connections
            { from: "401", to: "402" },
            { from: "402", to: "403" },
            { from: "403", to: "404" }
        ]
    };
    // Initialize variables
    let canvas, ctx;
    let currentFloor = 1;
    let path = [];
    let animationId = null;
    let camera = {
        x: 50,
        y: 50,
        z: 100,
        angle: 0,
        tilt: 30
    };

    window.onload = function() {
        canvas = document.getElementById('buildingCanvas');
        ctx = canvas.getContext('2d');

        resizeCanvas();
        window.addEventListener('resize', resizeCanvas);
```

```
            document.getElementById('findPath').addEventListener('click',
findPath);
    document.querySelectorAll('.floor-btn').forEach(btn => {
        btn.addEventListener('click', function() {
                        document.querySelectorAll('.floor-btn').forEach(b    =>
b.classList.remove('active'));
            this.classList.add('active');
            currentFloor = parseInt(this.dataset.floor);
            drawBuilding();
        });
    });
    drawBuilding();

    setupCameraControls();
};
function resizeCanvas() {
    canvas.width = canvas.clientWidth;
    canvas.height = canvas.clientHeight;
    drawBuilding();
}
function setupCameraControls() {
    let isDragging = false;
    let lastX, lastY;

    canvas.addEventListener('mousedown', function(e) {
        isDragging = true;
        lastX = e.clientX;
        lastY = e.clientY;
    });
    canvas.addEventListener('mousemove', function(e) {
        if (!isDragging) return;

        const deltaX = e.clientX - lastX;
        const deltaY = e.clientY - lastY;

        camera.angle += deltaX * 0.01;
          camera.tilt = Math.max(10, Math.min(80, camera.tilt + deltaY *
0.1));

        lastX = e.clientX;
        lastY = e.clientY;

        drawBuilding();
    });

    canvas.addEventListener('mouseup', function() {
        isDragging = false;
    });

    canvas.addEventListener('wheel', function(e) {
          camera.z = Math.max(50, Math.min(200, camera.z - e.deltaY *
0.1));
        drawBuilding();
        e.preventDefault();
    });
}

function drawBuilding() {
```

```
    ctx.clearRect(0, 0, canvas.width, canvas.height);

    ctx.save();
    ctx.translate(canvas.width / 2, canvas.height / 2);

    drawFloor();

    drawConnections();

    drawPoints();

    if (path.length > 0) {
        drawPath();
    }

    ctx.restore();
}
function drawFloor() {
    ctx.fillStyle = 'rgba(240, 240, 240, 0.8)';
    ctx.fillRect(-200, -100, 400, 200);
    ctx.strokeStyle = 'rgba(200, 200, 200, 0.5)';
    ctx.lineWidth = 0.5;

    for (let x = -200; x <= 200; x += 20) {
        ctx.beginPath();
        ctx.moveTo(x, -100);
        ctx.lineTo(x, 100);
        ctx.stroke();
    }
    for (let y = -100; y <= 100; y += 20) {
        ctx.beginPath();
        ctx.moveTo(-200, y);
        ctx.lineTo(200, y);
                                                ctx.stroke();
}
function drawConnectionLine(fromPoint, toPoint, isStairs = false) {
    const from = projectToCanvas(fromPoint);
    const to = projectToCanvas(toPoint);

    ctx.beginPath();
    ctx.moveTo(from.x, from.y);
    ctx.lineTo(to.x, to.y);

    if (isStairs) {
        ctx.strokeStyle = '#e74c3c';
        ctx.setLineDash([5, 3]);
    } else {
        ctx.strokeStyle = '#3498db';
        ctx.setLineDash([]);
    }
    ctx.stroke();
    ctx.setLineDash([]);
}
function drawStairsIndicator(point) {
    const projected = projectToCanvas(point);

    ctx.fillStyle = '#e74c3c';
    ctx.beginPath();
```

```
    ctx.arc(projected.x, projected.y, 5, 0, Math.PI * 2);
    ctx.fill();

    ctx.fillStyle = '#e74c3c';
    ctx.font = '12px Arial';
    ctx.fillText('Stairs', projected.x + 8, projected.y - 8);
}
function drawPoints() {
    Object.values(buildingData.points).forEach(point => {
        if (point.floor === currentFloor) {
            drawPoint(point);
        }
    });
}
function projectToCanvas(point) {
    const x3d = (point.x - 50) * 2;
    const y3d = (point.y - 50) * 2;
    const z3d = (point.floor - currentFloor) * 20;

    const cosAngle = Math.cos(camera.angle);
    const sinAngle = Math.sin(camera.angle);
    const cosTilt = Math.cos(camera.tilt * Math.PI / 180);
    const sinTilt = Math.sin(camera.tilt * Math.PI / 180);

    const xRotated = x3d * cosAngle - z3d * sinAngle;
    const zRotated = x3d * sinAngle + z3d * cosAngle;

    const yRotated = y3d * cosTilt - zRotated * sinTilt;
    const zFinal = y3d * sinTilt + zRotated * cosTilt;

    const scale = camera.z / (camera.z + zFinal);
    const x2d = xRotated * scale;
    const y2d = yRotated * scale;

    return { x: x2d, y: y2d };
}

function findPath() {
    const startId = document.getElementById('startPoint').value;
    const endId = document.getElementById('endPoint').value;

    if (startId === endId) {
        alert("Start and end points cannot be the same!");
        return;
    }

    path = bfs(startId, endId);

    if (path.length === 0) {
        alert("No path found!");
        return;
    }

    const distance = calculatePathDistance(path);
                document.getElementById('distance').textContent        =
distance.toFixed(1);

    const time = distance / 1.4 / 60; // in minutes
```

```
    document.getElementById('time').textContent = time.toFixed(1);

    // Switch to start point floor
    const startFloor = buildingData.points[startId].floor;
    currentFloor = startFloor;
    document.querySelectorAll('.floor-btn').forEach(btn => {
        btn.classList.remove('active');
        if (parseInt(btn.dataset.floor) === startFloor) {
            btn.classList.add('active');
        }
    });

    drawBuilding();

    startPathAnimation();
}

function bfs(startId, endId) {
    const queue = [[startId]];
    const visited = new Set([startId]);

    while (queue.length > 0) {
        const path = queue.shift();
        const node = path[path.length - 1];

        if (node === endId) {
            return path;
        }

        const neighbors = [];
        buildingData.connections.forEach(conn => {
            if (conn.from === node && !visited.has(conn.to)) {
                neighbors.push(conn.to);
                visited.add(conn.to);
            } else if (conn.to === node && !visited.has(conn.from)) {
                neighbors.push(conn.from);
                visited.add(conn.from);
            }
        });
        neighbors.forEach(neighbor => {
            queue.push([...path, neighbor]);
        });
    }
    return [];
}
function calculatePathDistance(path) {
    let distance = 0;

    for (let i = 0; i < path.length - 1; i++) {
        const fromPoint = buildingData.points[path[i]];
        const toPoint = buildingData.points[path[i+1]];
        const dx = fromPoint.x - toPoint.x;
        const dy = fromPoint.y - toPoint.y;
        const dz = (fromPoint.floor - toPoint.floor) * 3;

        distance += Math.sqrt(dx*dx + dy*dy + dz*dz);
    }
    return distance;
```

```
}
function startPathAnimation() {
    if (animationId) {
        cancelAnimationFrame(animationId);
    }
    let animationProgress = 0;
    const animationDuration = 5000;
    function animate() {
        animationProgress += 16;

        if (animationProgress >= animationDuration) {
            animationProgress = 0;
        }
        drawBuilding();
        drawAnimation(animationProgress / animationDuration);

        animationId = requestAnimationFrame(animate);
    }
    animate();
}
function drawAnimation(progress) {
    if (path.length < 2) return;

    const totalSegments = path.length - 1;
    const segmentProgress = progress * totalSegments;
    const segmentIndex = Math.floor(segmentProgress);
    const segmentFraction = segmentProgress - segmentIndex;

    if (segmentIndex >= totalSegments) return;

    const fromPoint = buildingData.points[path[segmentIndex]];
    const toPoint = buildingData.points[path[segmentIndex + 1]];

    if (fromPoint.floor === currentFloor && toPoint.floor === currentFloor)
{
            const currentX = fromPoint.x + (toPoint.x - fromPoint.x) *
segmentFraction;
            const currentY = fromPoint.y + (toPoint.y - fromPoint.y) *
segmentFraction;

        const currentPoint = {
            name: "Walking Path",
            x: currentX,
            y: currentY,
            floor: currentFloor
        };
        const projected = projectToCanvas(currentPoint);
        ctx.fillStyle = '#e74c3c';
        ctx.beginPath();
        ctx.arc(projected.x, projected.y, 8, 0, Math.PI * 2);
        ctx.fill();

        ctx.fillStyle = '#fff';
        ctx.beginPath();
        ctx.arc(projected.x, projected.y, 4, 0, Math.PI * 2);
        ctx.fill();
    }
}
```

B. Users Guide



Figure 10. Opening screen of the Mobile-Base 3D Navigation System

1. Wait until the progress bar reaches 100%, confirming that all 3D models, textures, and system components have been successfully loaded. Once the loading process is complete, the PATHFINDER button will become active. You may then tap or click it to enter the main interface of the application.

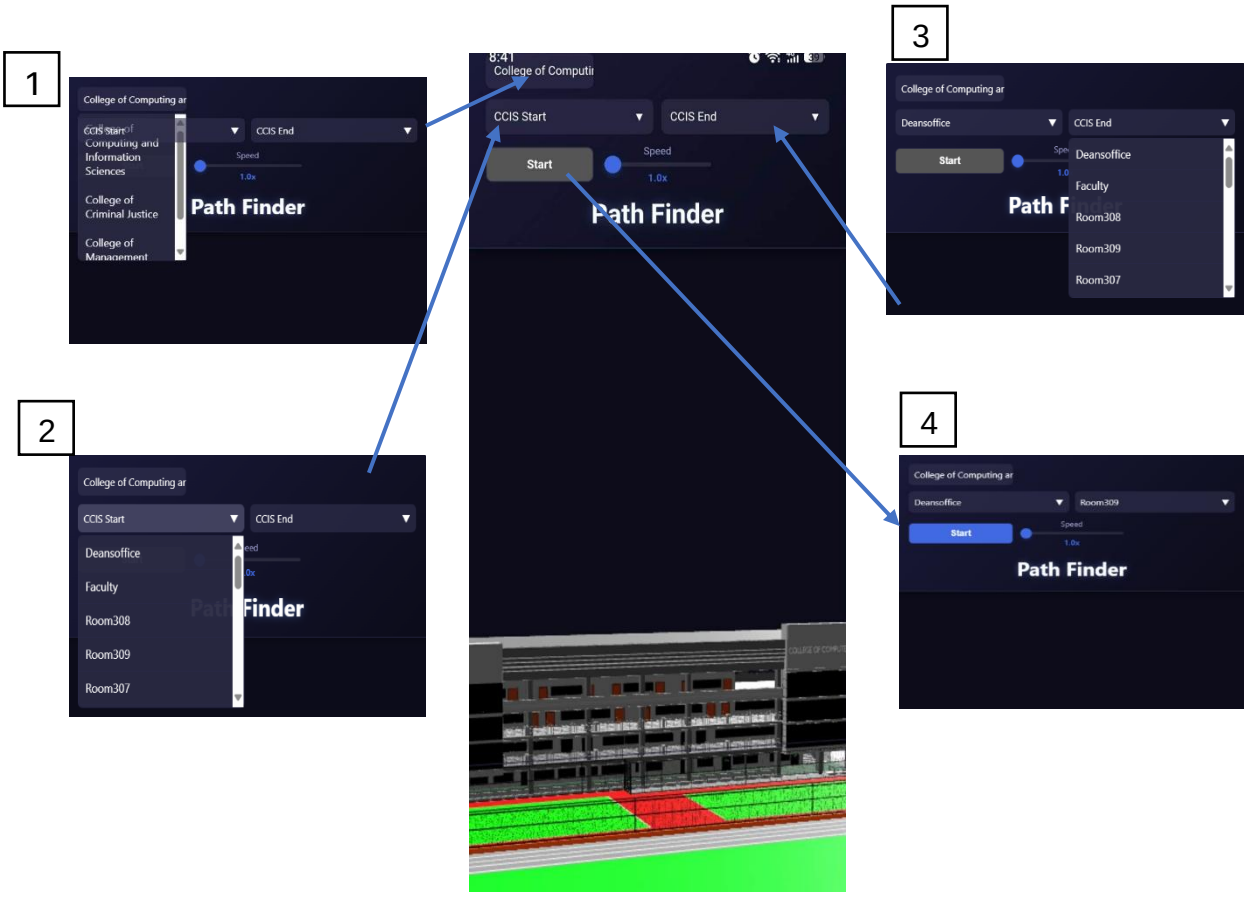


Figure 11. Search Bar Navigation Screen of the Mobile-Base 3D Navigation System

1. The user must first choose the college department they belong to. After selecting the department, the user then identifies their current location by choosing it from the available options. Once the current location is selected, the next step is to choose the desired destination. Finally, the user clicks the **Start** button to begin the navigation process.

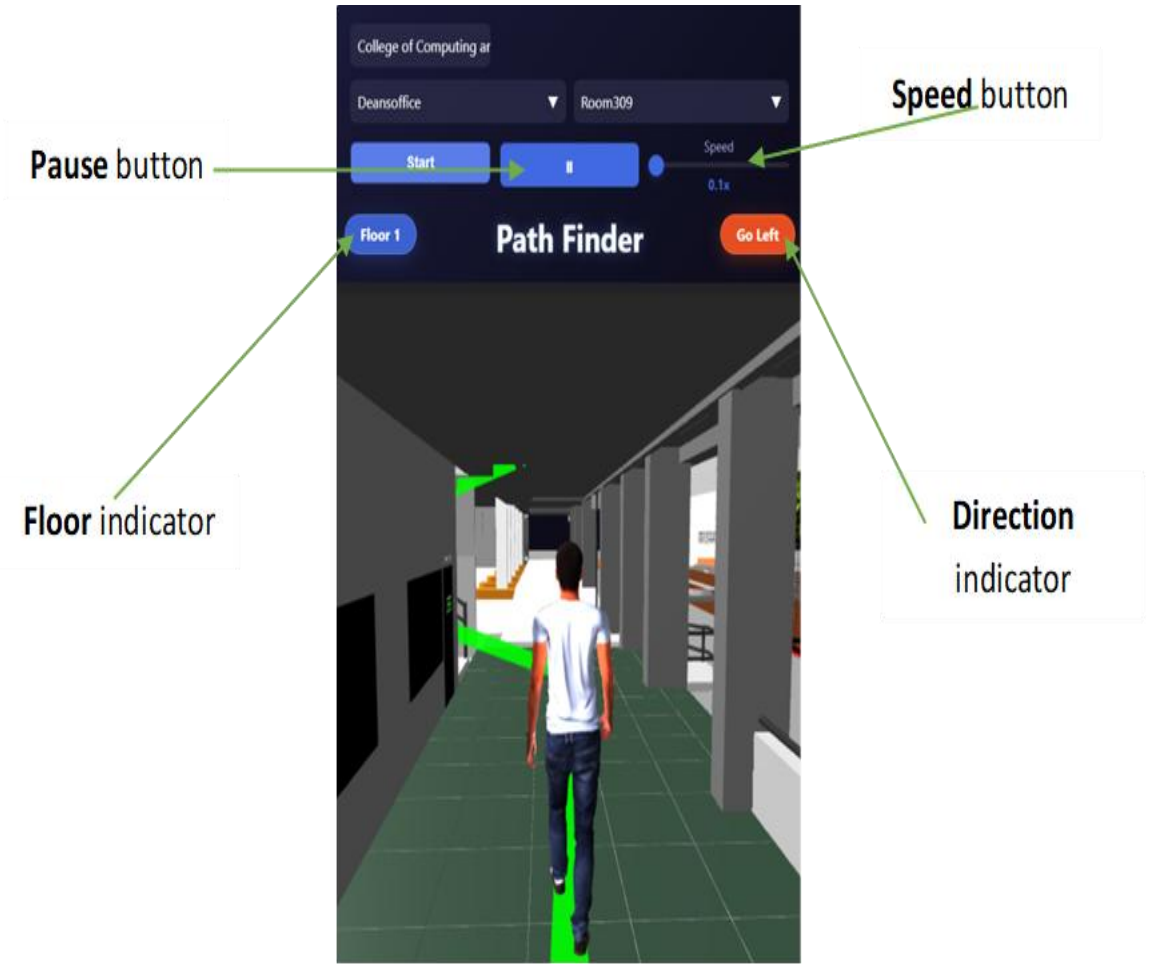


Figure 12. Pathway simulation Screen of the Mobile-Base 3D Navigation System

1. After the navigation starts, the user can access several control features to manage their movement along the 3D path. The user can pause the movement using the **Pause** button and adjust how fast the navigation progresses through the **Speed** button. The **Floor** indicator shows the current floor level, while the **Direction** indicator guides the user on where to turn or move next.

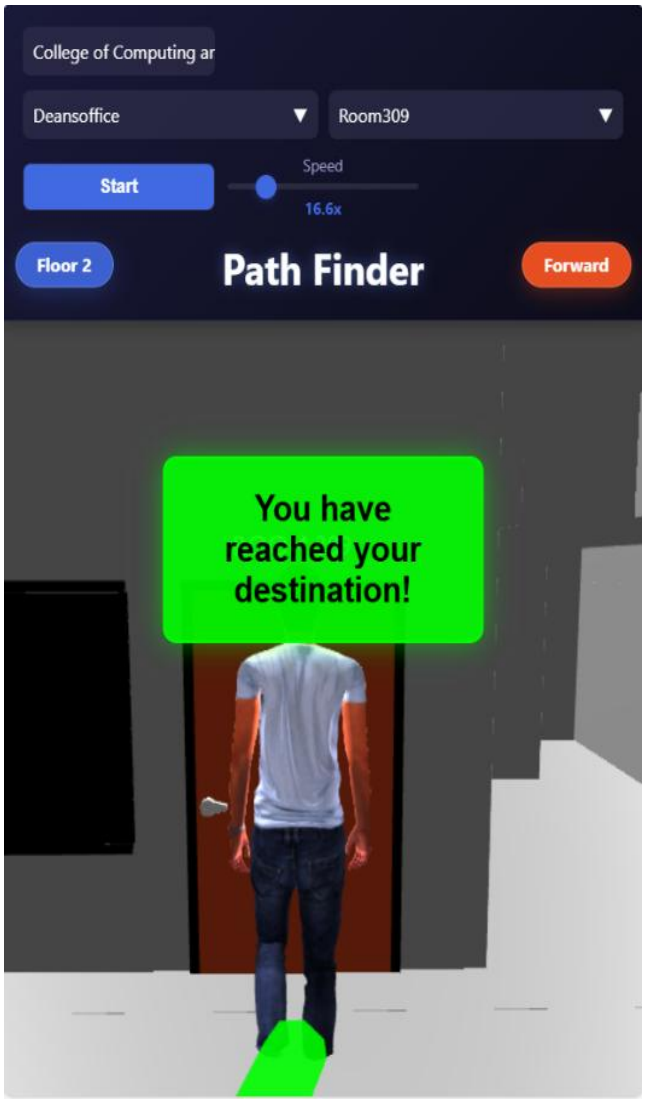


Figure 13. Destination Screen of the Simulated Path

1. Once the user reaches the selected location, a message appears on the screen stating “You have reached your destination!”. This serves as confirmation that the navigation has successfully ended.

F. Curriculum Vitae



JOHN ARNAIS M. RONDINA

Programmer/ Designer

PARENTS

- **NAME OF MOTHER:**
  Rosielyn R. Hilongo
- **NAME OF FATHER:**
  Arnais A. Hilongo

CONTACT ME

0965-865-2326

johnyrondina@gmail.com

PERSONAL

- DATE OF BIRTH: Oct 7, 2003
- NATIONALITY: Filipino
- RELIGION: Roman Catholic
- ADDRESS: Brgy.Panjobjoban2 Almagro, Samar

EDUCATION

- **PRIMARY:**
  Almagro Central Elementary School
  Panjobjoban, Almagro, Samar
  SY: 2009-2016
- **SECONDARY:**
  Almagro National High School
  Poblacion, Almagro, Samar
  SY: 2016-2022
- **TERTIARY:**
  Northwest Samar State University
  Ruesa St. Calbayog City, Samar
  Bachelor of Science in Information Technology
  SY: 2022-2026



ROBERT JOHN L, LABONITE

Progammer/ Designer

PERSONAL

- DATE OF BIRTH: Dec 26, 1997
- NATIONALITY: Filipino
- RELIGION: Roman Catholic
- ADDRESS: P3, Brgy. Carmen Calbayog City, Samar

EDUCATION

- **PRIMARY:**

  Carmen Elementary School
  P-2, Brgy. Carmen Calbayog City
  SY: 2005-2010

- **SECONDARY:**

  La Milagrosa Acdemy
  J.D. Avelino St. Calbayog City
  SY: 2010-2014

- **TERTIARY:**

  Northwest Samar State University
  Ruesa St. Calbayog City, Samar
  Bachelor of Science in Information Technology
  SY: 2022-2026

PARENTS

- **NAME OF MOTHER:**

  Liwayway Joan L. Labonite

- **NAME OF FATHER:**

  Roberto Y. Labonite Sr.

CONTACT ME

0970-061-1658

robertjohnlabonite267
@gmail.com



VIOSEL D. LOJERO

Researcher/ Document Specialist

PERSONAL

- DATE OF BIRTH: Sep 5, 2000
- NATIONALITY: Filipino
- RELIGION: Roman Catholic
- ADDRESS: Brgy. Migara Calbayog City, Samar

EDUCATION

- **PRIMARY:**

  San Jose Elementary School
  San Jose , calbayog city, samar
  SY: 2008-2013

- **SECONDARY:**

  Migara National High School
  Migara Calbayog City Samar
  SY: 2013-2020

- **TERTIARY:**

  Northwest Samar State University
  Ruesa St. Calbayog City, Samar
  Bachelor of Science in Information Technology
  SY: 2022-2026

PARENTS

- **NAME OF MOTHER:**
  Nimfa B. Abajo
- **NAME OF FATHER:**
  Levirato D. Lojero

CONTACT ME

0920-271-5660

lojeroviosel@gmail.com



JODELYN E. MONTERNEL

Researcher/ Document Specialist

PARENTS

- **NAME OF MOTHER:**
  Jocelyn E. Monternil
- **NAME OF FATHER:**
  Edelindo V. Monternil Sr.

CONTACT ME

0910-492-6678

jodelyn7283@gmail.com

PERSONAL

- DATE OF BIRTH: Jul 28, 2003
- NATIONALITY: Filipino
- RELIGION: Roman Catholic
- ADDRESS: P3, Brgy. Nabangig Palanas, Masbate

EDUCATION

- **PRIMARY:**
  Nabangig Central School
  Nabangig Palanas, masbate
  SY: 2009-2016
- **SECONDARY:**
  Rondina - Atendido National High School
  Nabangig Palanas, Masbate
  SY: 2016-2022
- **TERTIARY:**
  Northwest Samar State University
  Ruesa St. Calbayog City, Samar
  Bachelor of Science in Information Technology
  SY: 2022-2026



MARY ROSE E. LEBARIO

Project Leader/ Programmer/ Document Specialist

PERSONAL

- DATE OF BIRTH: Aug 12, 2001
- NATIONALITY: Filipino
- RELIGION: Roman Catholic
- ADDRESS: Brgy. Caparangasan Gandara, Samar

EDUCATION

- **PRIMARY:**
  Caparangasan Elementary School
  Caparangasan Gandara, Samar
  SY: 2009-2016
- **SECONDARY:**
  Pagsanghan National High School
  Pagsanghan , Samar
  SY: 2016-2022
- **TERTIARY:**
  Northwest Samar State University
  Ruesa St. Calbayog City, Samar
  Bachelor of Science in Information Technology
  SY: 2022-2026

PARENTS

- **NAME OF MOTHER:**
  Rosie C. Edalde
- **NAME OF FATHER:**
  Marcelo T. Lebario

CONTACT ME

0912-073-7480

lebariomaryrose1
@gmail.com